The Critical Coherence Index as a Projection Observable

Breadth–Depth Compression and Transition Robustness in a Minimal Cosmological Benchmark

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Abstract

The Critical Coherence Index (CCI) has been used throughout the structural selection programme as a reduced observable for structural stress and coherence loss. This paper examines a more specific interpretation: the CCI as a projection observable arising when high-dimensional latent structural depth is compressed onto an accessible cosmological surface.

We introduce a minimal breadth–depth projection model in which expansion generates configuration breadth, linear growth supplies accessible coherent depth, and a bounded projection residual defines a transition marker. A baseline LCDM-growth proxy benchmark yields a transition estimate $z_{\mathrm{tr}} \simeq 0.850$, while a sensitivity scan over 6930 parameter points gives a median transition redshift $z_{\mathrm{tr}} \simeq 0.889$ and a mean $z_{\mathrm{tr}} \simeq 0.845$.

The result is not a cosmological parameter fit and does not constitute evidence for modified gravity. Its contribution is methodological: it shows that the CCI can be operationally interpreted as a stabilized projection of expansion breadth against accessible structural depth, and that the resulting transition marker is reproducible across a broad toy-parameter grid.

1 Purpose and Status

Earlier papers introduced the cosmological CCI as a structural-stress observable combining expansion, growth, connectivity, and robustness sectors. Paper 32 then made first contact with Cosmic Chronometer $H(z)$ data. The present paper addresses a different question:

What kind of object is the CCI when it is used as a cosmological observable?

The answer proposed here is deliberately modest. The CCI is treated as a projection observable: a compressed measurement of the mismatch between expansion-driven breadth and accessible structural depth. This is not a new cosmological model, not a fit to data, and not a replacement for Boltzmann-code perturbation theory. It is a reproducible diagnostic benchmark designed to clarify the interpretation of the CCI.

2 Projection Interpretation

The guiding picture is that the universe contains more structural possibility than is directly accessible in a coarse-grained observable. Expansion increases the breadth of accessible configurations, while structure growth and coherence determine how much latent depth remains available.

We model this competition through three ingredients:

- **Breadth**: expansion-driven configuration volume,
- **Depth**: accessible coherent structure,
- **Projection**: bounded compression of latent structure into an observable scalar.

In this interpretation, a large CCI does not mean "large energy". It means that the projected surface of the system is increasingly broad relative to the coherent depth available to organise it.

3 Minimal LCDM-Growth Benchmark

The v0.3 benchmark uses a flat LCDM reference background with

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685. \tag{1}$$

The dimensionless expansion function is

$$E(z) = \sqrt{\Omega_{m0}(1+z)^3 + \Omega_{\Lambda 0}}. \tag{2}$$

The linear growth factor is approximated using the Carroll–Press–Turner growth-suppression proxy,

$$D(z) = \frac{g(z)/(1+z)}{g(0)}, \tag{3}$$

with

$$g(z) = \frac{5\Omega_m(z)/2}{\Omega_m(z)^{4/7} - \Omega_\Lambda(z) + (1+\Omega_m(z)/2)\,(1+\Omega_\Lambda(z)/70)}. \tag{4}$$

This is an analytic proxy, not a full perturbation solution.

3.1 Latent Structural Depth

Latent structural depth is represented by a smooth logistic sector,

$$L(z) = L_{\text{floor}} + (1 - L_{\text{floor}})\frac{1}{1+\exp[s(z-z_c)]}, \tag{5}$$

with baseline parameters

$$z_c = 1.1, \qquad s = 3.5, \qquad L_{\text{floor}} = 0.20. \tag{6}$$

The accessible depth is then

$$D_{\text{acc}}(z) = D(z)L(z). \tag{7}$$

3.2 Breadth and Projection CCI

Expansion breadth is defined as

$$B_{\text{exp}}(z) = E(z)(1+z). \tag{8}$$

The projection compression is bounded by a hyperbolic tangent,

$$Q_\alpha(z) = \tanh\left[\frac{(B_{\text{exp}}(z)/B_{\text{exp}}(0))^\alpha}{\max_z (B_{\text{exp}}(z)/B_{\text{exp}}(0))^\alpha}\right], \tag{9}$$

and the projected surface is

$$S_{\text{proj}}(z) = B_{\text{exp}}(z)Q_\alpha(z). \tag{10}$$

The projection CCI is

$$\boxed{\mathcal{C}_{\text{CCI}}^{\text{proj}}(z) = \frac{S_{\text{proj}}(z)}{1+\gamma D_{\text{acc}}(z)}} \tag{11}$$

and is normalized to $\mathcal{C}_{\text{CCI}}^{\text{proj}}(0) = 1$ for plotting.

4 Transition Estimator

The bounded projection residual is

$$R_{\mathrm{proj}}(z) = \frac{|S_{\mathrm{proj}}(z) - D_{\mathrm{acc}}(z)|}{S_{\mathrm{proj}}(z) + D_{\mathrm{acc}}(z) + \epsilon}. \tag{12}$$

The transition marker is estimated from the strongest residual curvature,

$$z_{\mathrm{tr}} = \underset{z \in [0.08 z_{\max},\, 0.92 z_{\max}]}{\arg\max} \left| \frac{\mathrm{d}^2 R_{\mathrm{proj}}}{\mathrm{d}z^2} \right|. \tag{13}$$

The edge exclusion prevents artificial boundary transitions.

5 Baseline Results

The baseline v0.3 benchmark gives

$$z_{\mathrm{tr}} = 0.84985, \qquad C_{\mathrm{norm}}(z_{\mathrm{tr}}) = 12.0466, \qquad R_{\mathrm{proj}}(z_{\mathrm{tr}}) = 3.21 \times 10^{-4}. \tag{14}$$

The final normalized projection CCI at $z = 3$ is

$$C_{\mathrm{norm}}(3) = 474.184. \tag{15}$$

The matter–Lambda equality redshift of the reference background is

$$z_{m\Lambda} = 0.29557. \tag{16}$$

Thus the baseline projection-transition marker is not identical to matter–Lambda equality; it occurs later in redshift and should be interpreted as a structural projection transition rather than a background density equality.

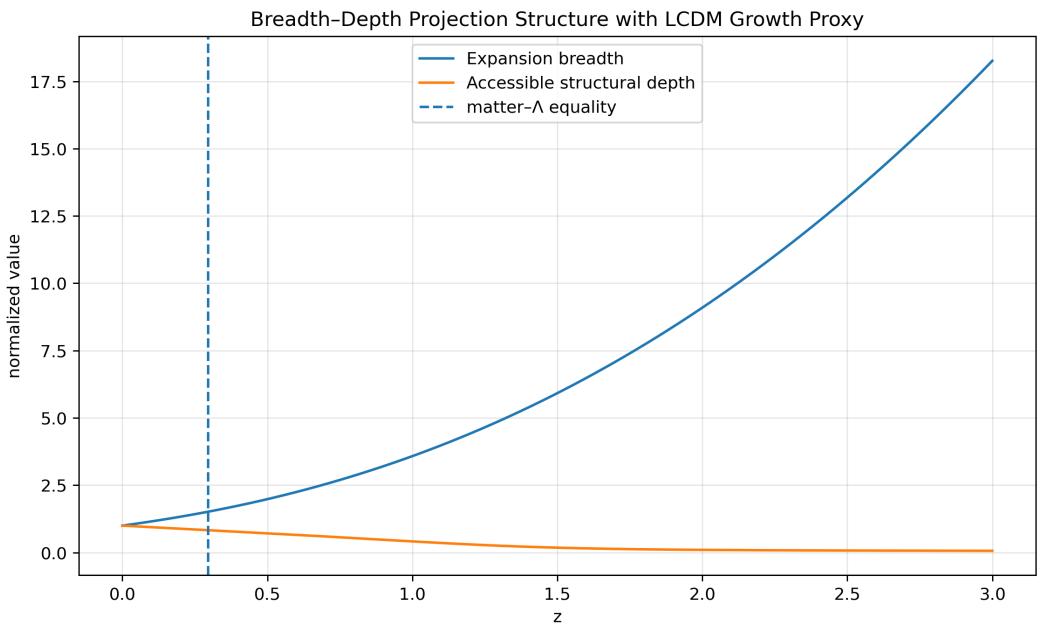

Figure 1: Expansion breadth and accessible structural depth in the v0.3 LCDM-growth proxy benchmark. The structural-depth channel decreases as redshift increases, while expansion breadth grows rapidly.

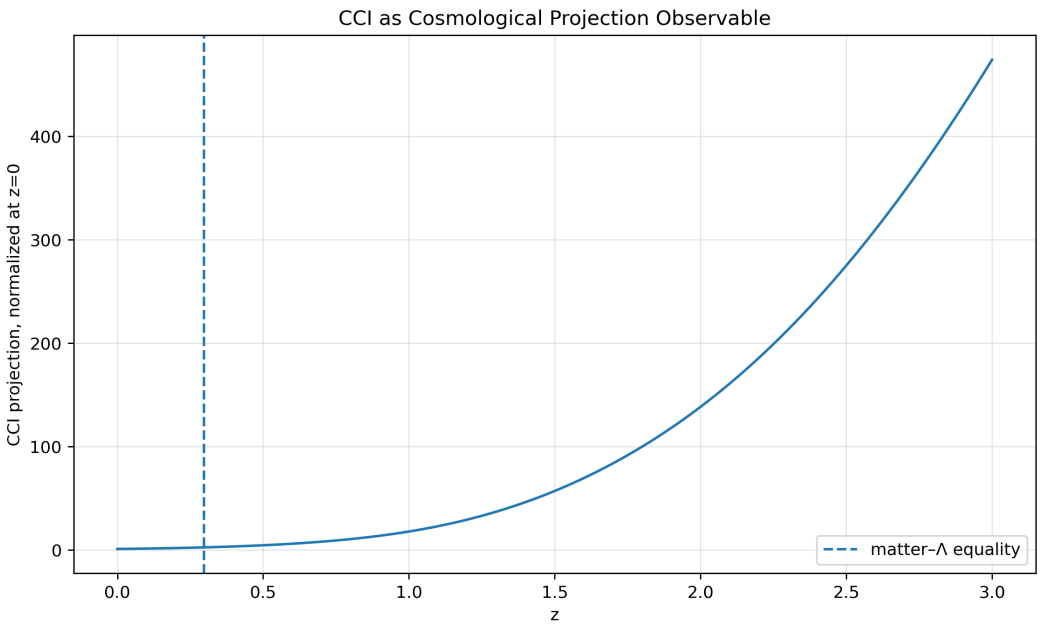

Figure 2: Normalized projection CCI. The observable grows as expansion breadth dominates accessible coherent depth.

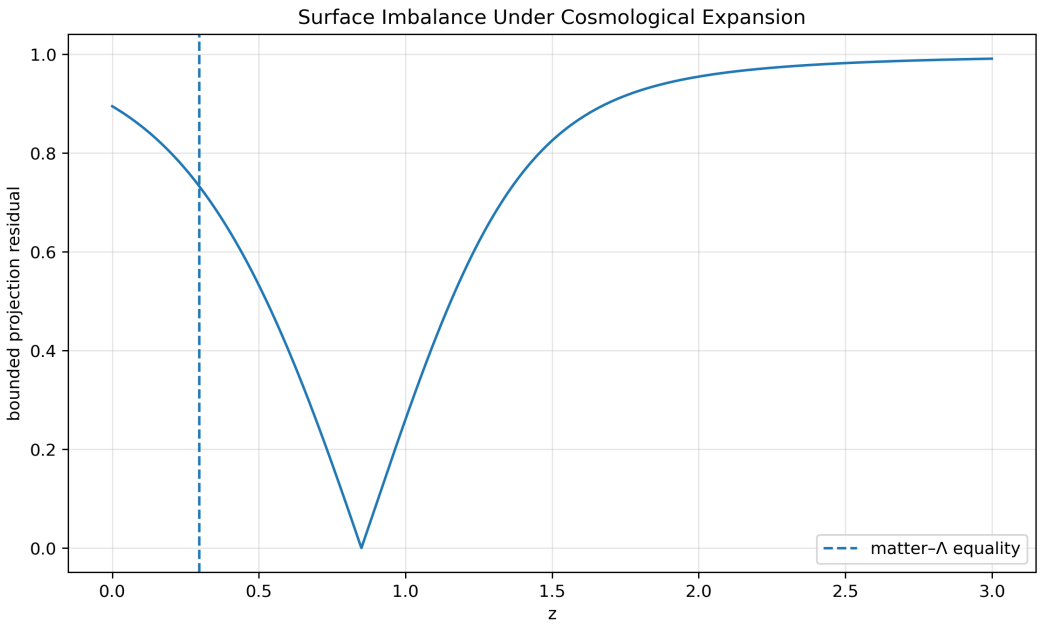

Figure 3: Bounded projection residual used to estimate the structural transition marker.

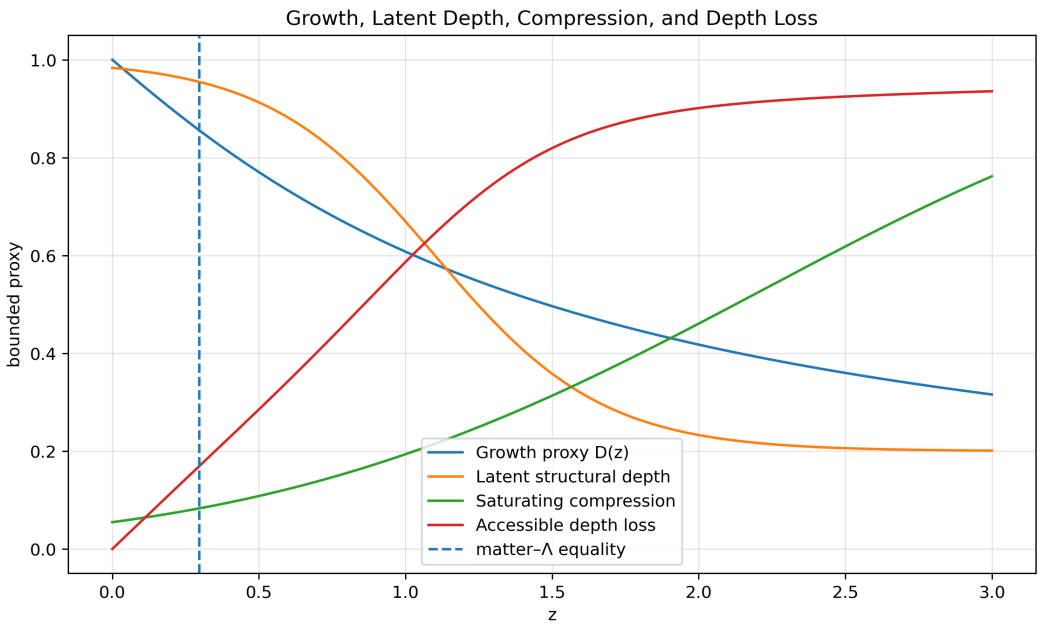

Figure 4: Component proxies entering the projection observable: growth, latent structural depth, saturating compression, and depth loss.

6 Sensitivity Scan

The v0.4 sensitivity test scans the parameter grid

$$\alpha \in [0.2, 2.0], \qquad z_c \in [0.6, 1.6], \tag{17}$$

$$s \in [2.0, 6.0], \qquad L_{\text{floor}} \in [0.10, 0.40], \tag{18}$$

for a total of 6930 parameter points.

Table 1: v0.4 sensitivity-scan summary.

Quantity	**Value**
Total parameter points	6930
Mean transition redshift	0.84491
Median transition redshift	0.88889
Standard deviation	0.30399
Minimum transition redshift	0.24024
Maximum transition redshift	1.38739
Fraction in $0.5 \leq z_{\text{tr}} \leq 1.1$	0.55657
Matter–Lambda equality redshift	0.29557

The scan shows that the projection transition is not a single tuned value. Instead, it occupies a broad but structured region, with a substantial concentration in the intermediate-redshift band.

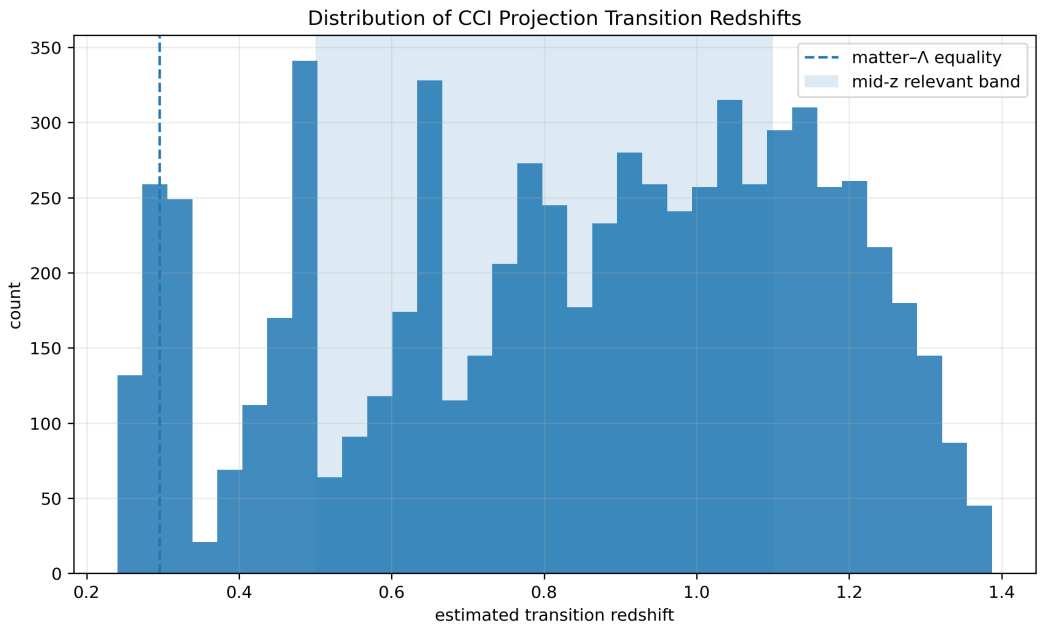

Figure 5: Distribution of transition redshifts across the v0.4 sensitivity grid. A large fraction lies in the intermediate-redshift band $0.5 \leq z \leq 1.1$.

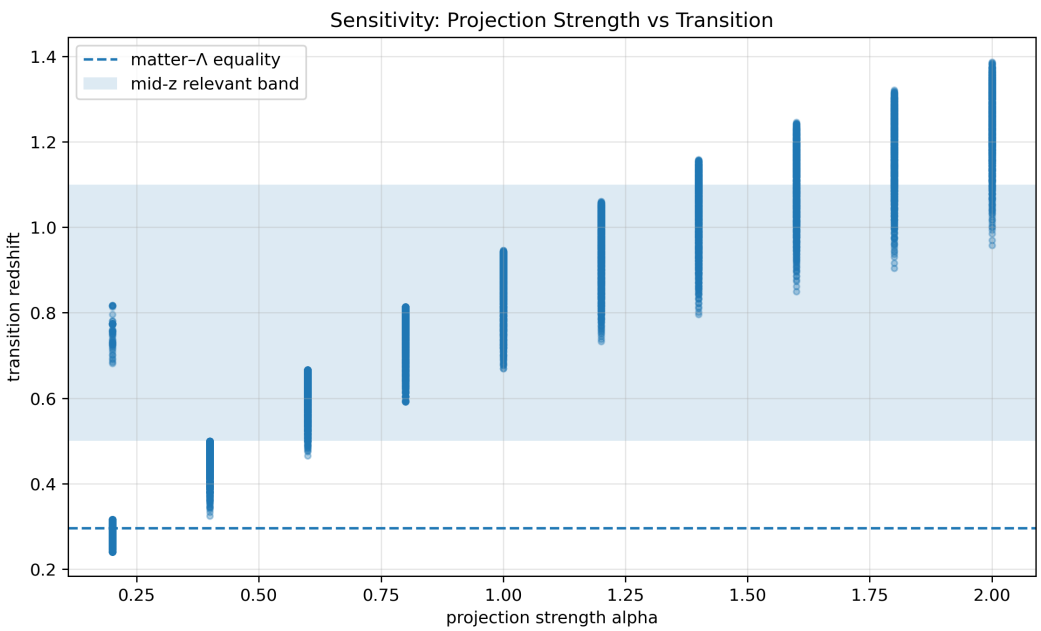

Figure 6: Projection strength α versus transition redshift. The transition location is sensitive to projection strength but remains structured rather than random across the grid.

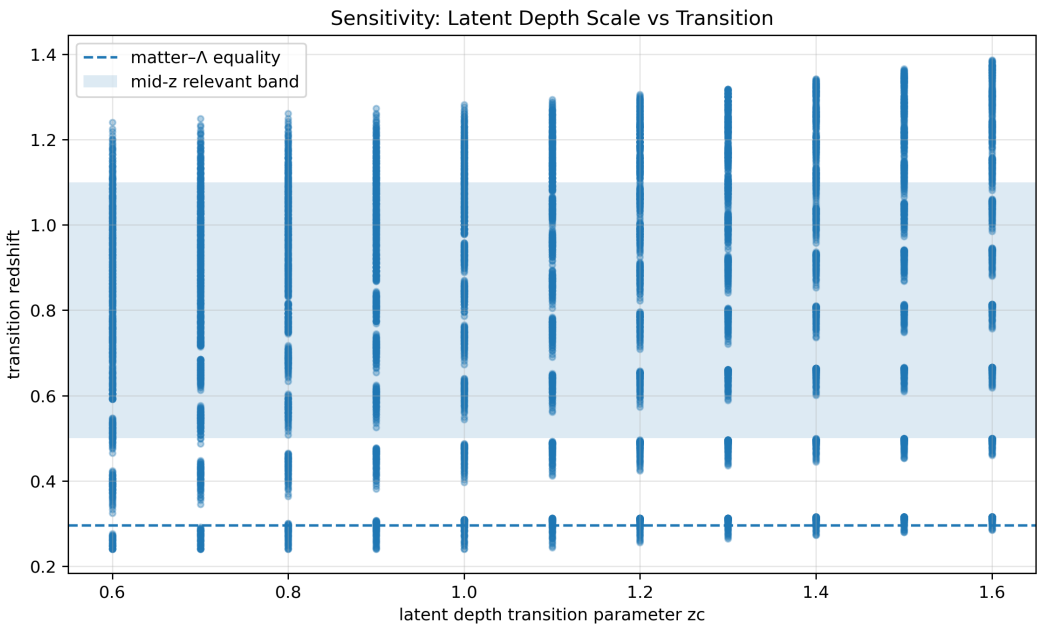

Figure 7: Latent-depth transition parameter z_c versus estimated projection-transition redshift.

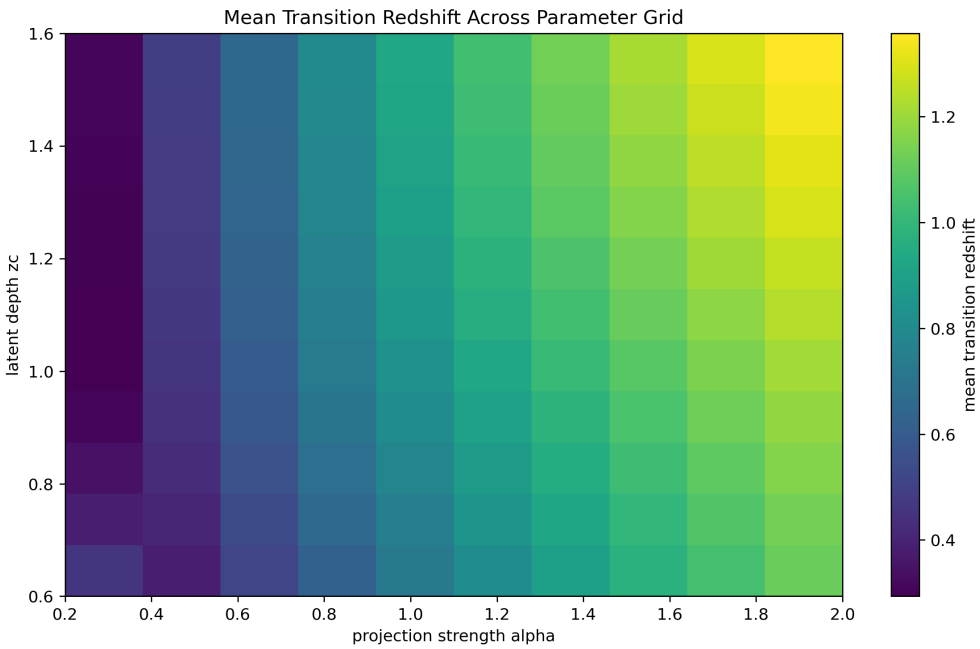

Figure 8: Mean transition redshift across the (α, z_c) projection grid, averaged over sharpness and depth-floor parameters.

7 Role of the Stabilized v0.2 Model

The v0.2 script is retained as a stabilization predecessor. It introduced three important numerical safeguards:

- a nonzero accessible-depth floor,
- saturating compression through tanh,

- transition estimation from residual curvature rather than boundary divergence.

In v0.2, the transition estimate is $z_{\mathrm{tr}} \simeq 0.454$ and the final normalized CCI at $z = 3$ is about 106.4. The later v0.3/v0.4 benchmarks should therefore be read as the physically improved LCDM-growth realisation of the stabilized projection idea.

8 Interpretation

The main interpretive result is:

> *The CCI can be read as a projection stress: the observable grows when expansion-driven breadth outpaces accessible coherent depth.*

This differs from an energy-only interpretation. The projection CCI is not defined by the absolute energy density, but by the relation between configuration breadth, structural depth, and compression. This is why the transition marker is not simply the matter–Lambda equality redshift.

The v04 sensitivity result also clarifies the status of the transition: it is not a finely tuned number, but a robust feature of a family of projection models. At the same time, its precise value is model-dependent and should not be treated as a measured cosmological transition without direct data fitting.

9 Limitations

The benchmark has several important limitations:

- The growth factor is analytic and approximate; no Boltzmann-code perturbation calculation is performed.
- The latent-depth sector is phenomenological and not derived from a microscopic cosmological field theory.
- The transition estimator is a diagnostic curvature marker, not an observed cosmological parameter.
- The sensitivity scan explores proxy parameters, not posterior cosmological constraints.
- No supernova, BAO, weak-lensing, or CMB likelihood is included.

10 Reproducibility

The scripts and generated artifacts are prepared for the public repository:

https://github.com/Chris4081/structural-selection-principle

Repository folder:

`experiments/maat_cci_projection_paper33/`

The main commands are:

```
python3 maat_cci_projection_test_v03.py
python3 maat_cci_projection_sensitivity_v04.py
```

The v0.2 stabilized predecessor is included for traceability:

```
python3 maat_cci_projection_test_v02.py
```

Each script writes CSV/JSON outputs and figures to its own output subfolder. The CSV and PNG files in the repository are derived analysis artifacts. The numerical background uses Planck-like LCDM parameter values and an analytic growth proxy; these external scientific inputs should be cited to their original sources when reused.

11 Conclusion

This paper reframes the CCI as a projection observable. In the minimal cosmological benchmark, expansion supplies breadth, linear growth and latent depth supply accessible coherence, and the CCI measures the resulting projection stress.

The baseline LCDM-growth proxy produces a transition marker near $z \simeq 0.85$, while the sensitivity scan shows that intermediate-redshift transitions persist across a broad parameter grid. The result is not a cosmological fit, but it provides a reproducible operational interpretation of the CCI as a stabilized breadth–depth compression observable.

References

[1] S. M. Carroll, W. H. Press, and E. L. Turner, "The Cosmological Constant," *Annual Review of Astronomy and Astrophysics* **30**, 499–542 (1992).

[2] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020).

[3] C. Krieg, "Cosmological Critical Coherence Index: A Structural-Stress Observable for Cosmic Evolution," preprint (2026).

[4] C. Krieg, "Cosmological Critical Coherence Index with Growth Connectivity and Robustness Margins," preprint (2026).

[5] C. Krieg, "Dynamic Structural Selection: From Response Fields to Observable Cosmology in the MAAT Framework," preprint (2026).

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